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How to Maintain a Lithium Battery for a Forklift: 5 Things That Can Make a Real Difference to Battery Life

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Lithium batteries are much easier to maintain than traditional lead-acid batteries.

There is no regular water topping-up, no electrolyte maintenance, and far less routine maintenance work for the operator.

But **“maintenance-free” does not mean “no maintenance required.”**

A forklift battery may go through frequent charging and discharging, high-current operation, opportunity charging, long working hours, vibration, and changing temperatures every day.

Even with high-quality lithium cells, the way the battery is used and stored can have a significant impact on its long-term performance.

In practice, good battery maintenance is less about doing more work and more about **avoiding the wrong conditions over a long period of time.**

Here are five areas that deserve particular attention.

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01 | A Battery That Is Not Being Used Still Needs Attention

One of the most common mistakes is surprisingly simple:

> The forklift is parked, so the battery is left alone.

Lithium batteries have relatively low self-discharge, but they are not completely inactive during storage.

A battery pack with a BMS, display, communication module, GPS, or other intelligent functions can still have a small amount of standby power consumption even when the forklift itself is not operating.

If the battery is already at a relatively low SOC when the forklift is parked, this gradual consumption can eventually bring the battery voltage down to a level where the BMS activates low-voltage protection.

That is why **long-term storage should include periodic checks rather than simply charging the battery once and leaving it for several months.**

There is also an important point that is sometimes misunderstood:

Long-term storage does not necessarily mean keeping the battery at 100% SOC all the time.

For most lithium-ion batteries, prolonged storage at very high SOC, particularly at elevated temperatures, can accelerate aging.

The appropriate storage SOC depends on the cell chemistry, battery design, BMS strategy, and the manufacturer's recommendations. In general, the battery should be stored within the recommended SOC range and checked periodically.

For a forklift that will remain unused for an extended period:

- Follow the battery manufacturer's recommended storage SOC.
- Check the SOC periodically rather than assuming the battery will remain unchanged.
- Recharge the battery when its SOC approaches the specified lower limit.
- Keep the battery in a dry, ventilated environment and avoid excessive heat.
- Before putting the forklift back into operation, recharge the battery according to the normal charging procedure.

For batteries equipped with an intelligent BMS, it is also useful to check the BMS status rather than relying only on the forklift's dashboard.

A battery that is not working is not necessarily a battery that is completely inactive.

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02 | Temperature Matters More Than Many Operators Realize

If SOC tells us the battery's state of charge, temperature tells us something equally important: **the conditions under which the cells are operating.**

For forklift lithium batteries, two situations deserve particular attention:

high-temperature charging and low-temperature charging.

At elevated temperatures, unwanted side reactions inside lithium-ion cells can accelerate. Prolonged exposure to high temperature, especially when combined with a high SOC, can contribute to faster cell aging.

Low temperatures create a different challenge.

As temperature drops, both charging and discharging performance can be affected. The issue becomes particularly important during charging.

For conventional lithium-ion cells with graphite-based anodes, charging at sufficiently low temperatures can lead to **lithium plating** if the charging conditions are not properly controlled.

This is not simply a matter of reduced performance. Lithium plating can cause irreversible capacity loss and may increase long-term reliability concerns.

This is why forklifts operating in cold storage facilities, outdoor environments, or winter conditions should not rely on a simple assumption:

> "If the charger starts, the battery can be charged."

A properly designed battery system may need to consider:

- Cell temperature monitoring
- BMS low-temperature charging protection
- Charging-current limitations based on temperature
- Battery heating when required
- Multiple temperature sensors for larger battery packs
- Appropriate charging strategies for cold environments

For applications such as cold-chain logistics and refrigerated warehouses, some lithium forklift batteries therefore incorporate **heating elements together with temperature sensing and BMS control logic.**

Temperature is not just another specification on a datasheet.

It is part of battery life management.

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03 | It Is Not Only About How Much You Charge — It Is About How You Charge

One of the major advantages of lithium batteries in forklifts is the ability to use **opportunity charging**.

Instead of waiting until the battery is nearly empty and then charging it for a long period, operators can recharge the battery during breaks or other available periods.

For warehouses, manufacturing facilities, and logistics operations with demanding schedules, this can significantly improve forklift utilization.

But opportunity charging does not mean:

> "Charge whenever you want, at any current you want."

The charging current, cutoff voltage, cell temperature, SOC, charger characteristics, and BMS strategy all need to work together.

Some practices should be avoided, including:

- Using a charger that is not properly matched to the battery
- Changing charging parameters without engineering validation
- Bypassing the BMS during charging
- Forcing the battery to charge outside its permitted temperature range
- Continuing to charge a battery that is already showing abnormal alarms or other signs of malfunction

A properly designed forklift battery system should therefore be viewed as a combination of:

Cells + BMS + Charger + Vehicle + Charging Strategy

rather than as a battery pack operating independently.

A good charging system is not necessarily the one that simply charges the fastest.

The real objective is to find the right balance between:

charging time, temperature rise, usable energy, operating requirements, and long-term battery life.



04 | Avoid Making “Almost Empty” the Normal Starting Point for Every Charge

There is still a common habit carried over from traditional lead-acid batteries:

> “Use the battery as much as possible, then charge it when it is almost empty.”

Lithium batteries are managed differently.

A BMS can protect the battery when voltage becomes too low, but **BMS low-voltage protection is a safety and protection function — it is not a target operating condition.**

Repeatedly operating a forklift until the battery reaches very low SOC can increase the likelihood of low-voltage protection events.

At low SOC, differences between individual cells can also become more noticeable, particularly when the battery is under a heavy load.

This is important for forklifts because traction loads can be highly dynamic.

A forklift may appear to have sufficient energy remaining, but when it suddenly accelerates, lifts a heavy load, or operates on an incline, the battery current can increase significantly.

At low SOC, this can result in a more pronounced voltage drop.

For this reason, a practical charging strategy should take into account:

- Battery capacity
- Forklift load profile
- Daily operating hours
- Charging opportunities
- BMS protection settings
- SOC measurement accuracy
- Actual operating conditions

The goal should not simply be to extract **every last percentage point of available energy** from the battery.

The goal is to keep the battery within a predictable operating range that matches the forklift's working requirements.

The BMS should be the last line of protection — not something the forklift reaches as part of its normal daily routine.

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05 | Battery Maintenance Also Means Checking the BMS and Electrical Connections

This is an area that is often overlooked in general battery-maintenance advice.

A forklift battery does not operate in a laboratory.

It experiences:

- Continuous vibration
- Repeated acceleration and braking
- High-current discharge
- Charging cycles
- Long operating hours
- Dust and moisture
- Mechanical stress around connectors and cables

So maintenance should not focus only on the cells.

The entire battery system should be checked periodically, including the **electrical connections and BMS condition.**

Depending on the battery design, routine inspections may include:

- Main positive and negative connections
- Power connectors
- Cable and terminal condition
- Signs of overheating or discoloration
- Loose connections
- Damaged or worn cables
- BMS alarms and protection events
- Cell-voltage consistency
- Temperature-sensor readings
- Unusual temperature differences between cells or areas of the pack
- Battery enclosure condition
- Any visible deformation or mechanical damage

If the BMS supports CAN, RS485, Bluetooth, or another communication interface, maintenance can go one step further.

Instead of waiting until the battery stops working, operating data can be reviewed to identify early signs of abnormal behavior.

For example, if one cell group repeatedly shows a voltage deviation from the others, or if one temperature sensor consistently reports an unusual value, that does not necessarily mean the battery has already failed.

But it **does mean the battery deserves further investigation.**

This is one of the biggest differences between simply “using a battery” and actually managing a battery system.

The Battery Is Only One Part of Battery Life

Forklift lithium battery life is not determined by the cells alone.

In real-world applications, it is influenced by a combination of:

**Cell chemistry

- BMS
- Charger
- Forklift operating conditions
- Temperature
- SOC management
- Charging habits
- Maintenance**

That is why I would be careful with the phrase:

> **"Lithium batteries are maintenance-free."**

A better way to describe them would be:

> **Lithium batteries eliminate much of the routine maintenance associated with lead-acid batteries, but they still require proper battery management.**

For a forklift operating 16–24 hours a day, relying heavily on opportunity charging, working in cold storage, or handling demanding loads, battery maintenance should not simply mean checking whether the battery still has enough energy.

It means understanding **how the battery is being charged, how deeply it is being discharged, what temperature it is operating at, how the BMS is responding, and whether the electrical system remains healthy.**

In the long run, these details are often what separate a battery that simply works from one that continues to work reliably year after year.

#ForkliftBattery #LithiumBattery #BatteryManagement #BatteryMaintenance #BMS #IndustrialMobility

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